Unit Test Report

Project: BookStore “Training Brain”
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Testing framework: JUnit 4 in NetBeans (Test Project, Alt+F6)
Execution date: October 28, 2026

1. Summary of Test Execution

This document lists the unit tests created and executed in NetBeans for the entities of the BookStore system. In total **166 tests** were executed: **150 passed**, **15 failed** and **1 ended with an error** (success rate 90.36 %).

Every failing or erroring test was written against the expected behavior of the system (business rules and validations that the code already applies elsewhere). Each one reveals a real defect in the source code and is listed in section 2 with its defect ID (DEF-xxx).

Entity / Class Tested	Tests Executed	Passed	Failed / Error	Execution Status
PaymentTest	12	10	2	FAILED
OrderTest	13	10	3	FAILED
AddressTest	10	10	0	PASSED
DeliveryDriverTest	10	10	0	PASSED
DeliveryAssignmentTest	12	10	2	FAILED
AuthorTest	10	10	0	PASSED
BookTest	14	10	4	FAILED
CategoryTest	10	10	0	PASSED
InventoryTest	11	10	1	FAILED
PublisherTest	10	10	0	PASSED
CustomerTest	11	10	1	FAILED
NotificationTest	10	10	0	PASSED
OrderDetailTest	10	10	0	PASSED
ReviewTest	12	10	2	FAILED

Entity / Class Tested	Tests Executed	Passed	Failed / Error	Execution Status
ShoppingCartTest	11	10	1	FAILED
TOTAL	**166**	**150**	**16**	**FAILED**

2. Defects Found by the Tests

Defect ID	Test	Defect description	Expected	Actual result	State
DEF-BOK-01	BookTest.testConstructorRejectsNegativeStock()	Creates a Book with stock -5 (setStock() rejects negatives, the constructor does not)	Throws IllegalArgumentException	Book is created with negative stock	Open
DEF-BOK-02	BookTest.testSetPriceNegativeIsRejected()	Sets a negative price (-5.0f)	Throws IllegalArgumentException	Negative price is silently ignored, no error reported	Open
DEF-BOK-03	BookTest.testUpdateStockBelowZeroIsRejected()	Removes 20 units from a stock of 10	Throws IllegalArgumentException	Operation is silently ignored	Open
DEF-BOK-04	BookTest.testSetAuthorKeepsAuthorBookListsInSync()	Changes the author of a book and checks both authors' book lists	New author lists the book, old author does not	Author lists are not updated	Open
DEF-CUS-01	CustomerTest.testSetEmailInvalidFormatIsRejected()	Sets the email "not-an-email" (age and password are validated, email is not)	Throws IllegalArgumentException	Invalid email is accepted	Open
DEF-ASN-01	DeliveryAssignmentTest.testAssignNullDriverDoesNotMarkAsAssigned()	Assigns a null driver to a delivery	Status is not "Assigned" (or exception)	Status changes to "Assigned"	Open
DEF-ASN-02	DeliveryAssignmentTest.testCompleteDeliveryWithoutDriverIsNotAllowed()	Completes a delivery that has no driver	Status is not "Completed" (or exception)	Status changes to "Completed"	Open
DEF-INV-01	InventoryTest.testSetReorderLevelNegativeIsRejected()	Sets a negative reorder level (-3)	Throws IllegalArgumentException	Negative level is accepted	Open
DEF-ORD-01	OrderTest.testCancelDeliveredOrderIsNotAllowed()	Cancels an order whose status is "Delivered"	Status stays "Delivered" (or exception)	Status changes to "Cancelled"	Open
DEF-ORD-02	OrderTest.testConfirmEmptyOrderIsNotAllowed()	Confirms an order without any detail	Status stays "Pending" (or exception)	Status changes to "Confirmed"	Open
DEF-ORD-03	OrderTest.testSetTotalNegativeIsRejected()	Sets a negative total (-50.0f)	Throws IllegalArgumentException	Negative total is accepted	Open
DEF-PAY-01	PaymentTest.testProcessInvalidPaymentIsNotProcessed()	Processes a payment with amount 0 and empty reference (validatePayment() is false)	Status stays "Pending" (or exception)	Status changes to "Processed"	Open
DEF-PAY-02	PaymentTest.testSetAmountNegativeIsRejected()	Sets a negative amount (-10.0f)	Throws IllegalArgumentException	Negative amount is accepted	Open
DEF-REV-01	ReviewTest.testSetScoreOutOfRangeIsRejected()	Sets a score of 9 (valid range 1-5)	Throws IllegalArgumentException	Out-of-range score is accepted	Open

Defect ID	Test	Defect description	Expected	Actual result	State
DEF-REV-02	ReviewTest.testPostSameReviewTwiceDoesNotDuplicate()	Posts the same review twice into the list	List size is 1	List size is 2 (duplicate)	Open
DEF-CRT-01	ShoppingCartTest.testRemoveNullBookReturnsFalse()	Removes null from a cart that has one item	Returns false	NullPointerException	Open

3. Test Cases Specification

3.1. PaymentTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-PAY-01	validatePayment()	Verifies: validate payment	Returns true	PASSED
TC-PAY-02	getPaymentStatus()	Getter returns the initialized value	Returns “Completed”	PASSED
TC-PAY-03	getId()	Getter returns the initialized value	Returns 100	PASSED
TC-PAY-04	getAmount()	Getter returns the initialized value	Returns 89.99f	PASSED
TC-PAY-05	setId()	Setter stores the new value	Returns 500	PASSED
TC-PAY-06	setAmount()	Setter stores the new value	Returns 150.0f	PASSED
TC-PAY-07	getPaymentMethod()	Getter returns the initialized value	Returns “Transferencia”	PASSED
TC-PAY-08	setStatus()	Setter stores the new value	Returns “Rejected”	PASSED
TC-PAY-09	getReferenceNumber()	Getter returns the initialized value	Returns “REF-199”	PASSED
TC-PAY-10	setReferenceNumber()	Setter stores the new value	Returns “REF-777”	PASSED
TC-PAY-11	processInvalidPaymentIsNotProcessed()	Processes a payment with amount 0 and empty reference (validatePayment() is false)	Status stays “Pending” (or exception)	FAILED (DEF-PAY-01)
TC-PAY-12	setAmountNegativeIsRejected()	Sets a negative amount (-10.0f)	Throws IllegalArgumentException	FAILED (DEF-PAY-02)

3.2. OrderTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-ORD-01	confirmOrder()	Confirms an order that has details	Status equals “Confirmed”	PASSED
TC-ORD-02	calculateTotal()	Adds a detail (2 units x 75.25) and calculates the total	Returns 150.50f	PASSED
TC-ORD-03	getId()	Getter returns the initialized value	Returns 10	PASSED
TC-ORD-04	getPaymentMethod()	Getter returns the initialized value	Returns “Transferencia”	PASSED
TC-ORD-05	cancelOrder()	Cancels a pending order	Status equals “Cancelled”	PASSED
TC-ORD-06	updateStatus()	Verifies: update status	Returns “Shipped”	PASSED
TC-ORD-07	setId()	Setter stores the new value	Returns 100	PASSED
TC-ORD-08	setTotal()	Setter stores the new value	Returns 90.0f	PASSED
TC-ORD-09	setPaymentMethod()	Setter stores the new value	Returns “Paypal”	PASSED
TC-ORD-10	toString()	Checks the text returned by toString()	Returns “Order{id=9, total=60.0, status=Pending}”	PASSED
TC-ORD-11	cancelDeliveredOrderIsNotAllowed()	Cancels an order whose status is “Delivered”	Status stays “Delivered” (or exception)	FAILED (DEF-ORD-01)

Test ID	Method Tested	Description	Expected Result	Status
TC-ORD-12	confirmEmptyOrderIsNotAllowed()	Confirms an order without any detail	Status stays “Pending” (or exception)	FAILED (DEF-ORD-02)
TC-ORD-13	setTotalNegativeIsRejected()	Sets a negative total (-50.0f)	Throws IllegalArgumentException	FAILED (DEF-ORD-03)

3.3. AddressTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-ADD-01	getFormattedAddress()	Formats the address	Returns the formatted address text	PASSED
TC-ADD-02	getMainStreet()	Getter returns the initialized value	Returns “Calle Caldas”	PASSED
TC-ADD-03	getSector()	Getter returns the initialized value	Returns “Sur”	PASSED
TC-ADD-04	getCoordinates()	Getter returns the initialized value	Object is not null	PASSED
TC-ADD-05	getSecondaryStreet()	Getter returns the initialized value	Returns “10 de Agosto”	PASSED
TC-ADD-06	getReference()	Getter returns the initialized value	Returns “Edificio Azul”	PASSED

Test ID	Method Tested	Description	Expected Result	Status
TC-ADD-07	setMainStreet()	Setter stores the new value	Returns “Calle Principal”	PASSED
TC-ADD-08	setSecondaryStreet()	Setter stores the new value	Returns “Calle Secundaria”	PASSED
TC-ADD-09	setSector()	Setter stores the new value	Returns “Norte”	PASSED
TC-ADD-10	toString()	Checks the text returned by toString()	Returns “Address{mainStreet=Av. Napo, sector=Sur}”	PASSED

3.4. DeliveryDriverTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-DRV-01	calculateCost()	Verifies: calculate cost	Returns 5.0f	PASSED
TC-DRV-02	updateStatus()	Verifies: update status	Returns “On Duty”	PASSED
TC-DRV-03	getId()	Getter returns the initialized value	Returns “D-03”	PASSED
TC-DRV-04	seeLocation()	Verifies: see location	Object is not null	PASSED
TC-DRV-05	getName()	Getter returns the initialized value	Returns “Pedro Picapiedra”	PASSED
TC-DRV-06	getVehiclePlate()	Getter returns the initialized value	Returns “PBF-1122”	PASSED
TC-DRV-07	setId()	Setter stores the new value	Returns “D-100”	PASSED
TC-DRV-08	setPhone()	Setter stores the new value	Returns “0999999999”	PASSED

Test ID	Method Tested	Description	Expected Result	Status
TC-DRV-09	setStatus()	Setter stores the new value	Returns “Offline”	PASSED
TC-DRV-10	toString()	Checks the text returned by toString()	Returns the formatted text	PASSED

3.5. DeliveryAssignmentTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-ASN-01	assignDriver()	Verifies: assign driver	Returns “Assigned”	PASSED
TC-ASN-02	completeDelivery()	Verifies: complete delivery	Returns “Completed”	PASSED
TC-ASN-03	getId()	Getter returns the initialized value	Returns 100	PASSED
TC-ASN-04	getDispatchTime()	Gets the dispatch date	Returns the assigned Date object	PASSED
TC-ASN-05	getStatus()	Getter returns the initialized value	Returns “In Transit”	PASSED
TC-ASN-06	setId()	Setter stores the new value	Returns 200	PASSED
TC-ASN-07	setStatus()	Setter stores the new value	Returns “Cancelled”	PASSED
TC-ASN-08	getDriver()	Getter returns the initialized value	Object is not null	PASSED
TC-ASN-09	getOrder()	Getter returns the initialized value	Object is not null	PASSED
TC-ASN-10	toString()	Checks the text returned by toString()	Returns the formatted text	PASSED

Test ID	Method Tested	Description	Expected Result	Status
TC-ASN-11	assignNullDriverDoesNotMarkAsAssigned()	Assigns a null driver to a delivery	Status is not “Assigned” (or exception)	FAILED (DEF-ASN-01)
TC-ASN-12	completeDeliveryWithoutDriverIsNotAllowed()	Completes a delivery that has no driver	Status is not “Completed” (or exception)	FAILED (DEF-ASN-02)

3.6. AuthorTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-AUT-01	getId()	Getter returns the initialized value	Returns 1	PASSED
TC-AUT-02	getName()	Getter returns the initialized value	Returns “Mario Vargas Llosa”	PASSED
TC-AUT-03	getNationality()	Getter returns the initialized value	Returns “Argentina”	PASSED
TC-AUT-04	getBiography()	Getter returns the initialized value	Returns “Autora de La casa de los espiritus”	PASSED
TC-AUT-05	getBooks()	Getter returns the initialized value	Object is not null	PASSED
TC-AUT-06	setId()	Setter stores the new value	Returns 500	PASSED
TC-AUT-07	setName()	Setter stores the new value	Returns “Pablo Neruda”	PASSED
TC-AUT-08	setNationality()	Setter stores the new value	Returns “Española”	PASSED
TC-AUT-09	setBiography()	Setter stores the new value	Returns “Novelista del siglo XX”	PASSED
TC-AUT-10	addBook()	Adds 1 book to the author (no duplicates)	Books list size is 1	PASSED

3.7. BookTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-BOK-01	getId()	Getter returns the initialized value	Returns 1	PASSED
TC-BOK-02	getTitle()	Getter returns the initialized value	Returns “Don Quijote de la Mancha”	PASSED
TC-BOK-03	checkAvailability()	Verifies: check availability	Returns true	PASSED
TC-BOK-04	getPrice()	Getter returns the initialized value	Returns 22.99f	PASSED
TC-BOK-05	setId()	Setter stores the new value	Returns 500	PASSED
TC-BOK-06	setTitle()	Setter stores the new value	Returns “Ficciones”	PASSED
TC-BOK-07	setIsbn()	Setter stores the new value	Returns “978-0000000000”	PASSED
TC-BOK-08	setPrice()	Setter stores the new value	Returns 18.50f	PASSED
TC-BOK-09	updateStock()	Updates the stock by +5 (initial stock 10)	Returns 15	PASSED
TC-BOK-10	getInfo()	Getter returns the initialized value	Returns the formatted text	PASSED

Test ID	Method Tested	Description	Expected Result	Status
TC-BOK-11	constructorRejectsNegativeStock()	Creates a Book with stock -5 (setStock() rejects negatives, the constructor does not)	Throws IllegalArgumentException	FAILED (DEF-BOK-01)
TC-BOK-12	setPriceNegativeIsRejected()	Sets a negative price (-5.0f)	Throws IllegalArgumentException	FAILED (DEF-BOK-02)
TC-BOK-13	updateStockBelowZeroIsRejected()	Removes 20 units from a stock of 10	Throws IllegalArgumentException	FAILED (DEF-BOK-03)
TC-BOK-14	setAuthorKeepsAuthorBookListsInSync()	Changes the author of a book and checks both authors' book lists	New author lists the book, old author does not	FAILED (DEF-BOK-04)

3.8. CategoryTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-CAT-01	getId()	Getter returns the initialized value	Returns 1	PASSED
TC-CAT-02	getName()	Getter returns the initialized value	Returns “Historia”	PASSED
TC-CAT-03	getDescription()	Getter returns the initialized value	Returns “Textos divulgativos y cientificos”	PASSED
TC-CAT-04	getBooks()	Getter returns the initialized value	Object is not null	PASSED

Test ID	Method Tested	Description	Expected Result	Status
TC-CAT-05	getBooksSize()	Getter returns the initialized value	Returns 0	PASSED
TC-CAT-06	toString()	Checks the text returned by toString()	Returns “Category{id=6, name=Poesia}”	PASSED
TC-CAT-07	addBooksNull()	Verifies: add books null	Returns 0	PASSED
TC-CAT-08	setId()	Setter stores the new value	Returns 500	PASSED
TC-CAT-09	setName()	Setter stores the new value	Returns “Biografia”	PASSED
TC-CAT-10	setDescription()	Setter stores the new value	Returns “Estudios de la materia”	PASSED

3.9. InventoryTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-INV-01	getReorderLevel()	Getter returns the initialized value	Returns 5	PASSED
TC-INV-02	getLastRestockDate()	Gets the last restock date	Returns the initial restock date	PASSED
TC-INV-03	getBook()	Gets the associated book	Returns the associated Book instance	PASSED
TC-INV-04	getBookNotNull()	Getter returns the initialized value	Object is not null	PASSED

Test ID	Method Tested	Description	Expected Result	Status
TC-INV-05	checkReorderStatus()	Checks reorder status with stock 15 and reorder level 5	Returns false (no reorder needed)	PASSED
TC-INV-06	setReorderLevel()	Setter stores the new value	Returns 10	PASSED
TC-INV-07	setLastRestockDate()	Sets a new restock date and reads it back	Returns the new date	PASSED
TC-INV-08	setBook()	Replaces the associated book and reads it back	Returns the new Book instance	PASSED
TC-INV-09	updateStock()	Updates the book stock by +5 (initial stock 10)	Book stock is 15	PASSED
TC-INV-10	toString()	Checks the text returned by toString()	Returns the formatted inventory text	PASSED
TC-INV-11	setReorderLevelNegativeIsRejected()	Sets a negative reorder level (-3)	Throws IllegalArgumentException	FAILED (DEF-INV-01)

3.10. PublisherTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-PUB-01	getId()	Getter returns the initialized value	Returns 1	PASSED
TC-PUB-02	getName()	Getter returns the initialized value	Returns “Penguin Random House”	PASSED
TC-PUB-03	getDescription()	Getter returns the initialized value	Returns “Textos educativos”	PASSED

Test ID	Method Tested	Description	Expected Result	Status
TC-PUB-04	getBooks()	Getter returns the initialized value	Object is not null	PASSED
TC-PUB-05	toString()	Checks the text returned by toString()	Returns “Publisher{id=5, name=Norma}”	PASSED
TC-PUB-06	setId()	Setter stores the new value	Returns 500	PASSED
TC-PUB-07	setName()	Setter stores the new value	Returns “Pearson”	PASSED
TC-PUB-08	setDescription()	Setter stores the new value	Returns “Libros de ingenieria”	PASSED
TC-PUB-09	viewBooks()	Creates 3 books for the publisher and calls viewBooks()	Publisher holds 3 books	PASSED
TC-PUB-10	addBook()	Adds 1 book to the publisher	Books list size is 1	PASSED

3.11. CustomerTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-CUS-01	register()	Verifies: register	Returns true	PASSED
TC-CUS-02	registerDuplicateEmail()	Verifies: register duplicate email	Returns false	PASSED
TC-CUS-03	loginCorrectCredentials()	Verifies: login correct credentials	Returns the same object	PASSED

Test ID	Method Tested	Description	Expected Result	Status
TC-CUS-04	loginWrongPassword()	Verifies: login wrong password	Returns null	PASSED
TC-CUS-05	loginNullEmail()	Verifies: login null email	Returns null	PASSED
TC-CUS-06	setAgeInvalid()	Invalid input is rejected	Throws IllegalArgumentException	PASSED
TC-CUS-07	setPassword()	Setter stores the new value	Returns “newpass123”	PASSED
TC-CUS-08	getName()	Getter returns the initialized value	Returns “Ana Perez”	PASSED
TC-CUS-09	setAge()	Setter stores the new value	Returns 30	PASSED
TC-CUS-10	loginUnknownEmail()	Verifies: login unknown email	Returns null	PASSED
TC-CUS-11	setEmailInvalidFormatIsRejected()	Sets the email “not-an-email” (age and password are validated, email is not)	Throws IllegalArgumentException	FAILED (DEF-CUS-01)

3.12. NotificationTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-NOT-01	sendNotification()	Verifies: send notification	Object is not null	PASSED
TC-NOT-02	sendNotificationSetsUnread()	Verifies: send notification sets unread	Returns false	PASSED

Test ID	Method Tested	Description	Expected Result	Status
TC-NOT-03	markAsRead()	Verifies: mark as read	Returns true	PASSED
TC-NOT-04	getMessage()	Getter returns the initialized value	Returns “Your order was shipped”	PASSED
TC-NOT-05	getCustomer()	Getter returns the initialized value	Returns the same object	PASSED
TC-NOT-06	setId()	Setter stores the new value	Returns 20	PASSED
TC-NOT-07	setMessage()	Setter stores the new value	Returns “Payment received”	PASSED
TC-NOT-08	isReadAfterMarkAsRead()	Verifies: is read after mark as read	Returns true	PASSED
TC-NOT-09	setRead()	Setter stores the new value	Returns true	PASSED
TC-NOT-10	toString()	Checks the text returned by toString()	Returns the formatted text	PASSED

3.13. OrderDetailTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-ODT-01	calculateSubTotal()	Verifies: calculate sub total	Returns 20.0f	PASSED
TC-ODT-02	updateQuantity()	Updates the quantity to 5	getQuantity() returns 5	PASSED

Test ID	Method Tested	Description	Expected Result	Status
TC-ODT-03	updateQuantityRecalculatesSubTotal()	Verifies: update quantity recalculates sub total	Returns 50.0f	PASSED
TC-ODT-04	updateQuantityInvalid()	Invalid input is rejected	Throws IllegalArgumentException	PASSED
TC-ODT-05	setUnitPriceNegative()	Invalid input is rejected	Throws IllegalArgumentException	PASSED
TC-ODT-06	getBook()	Getter returns the initialized value	Returns the same object	PASSED
TC-ODT-07	getQuantity()	Getter returns the initialized value	Returns 2	PASSED
TC-ODT-08	setQuantitySubTotal()	Setter stores the new value	Returns 40.0f	PASSED
TC-ODT-09	setUnitPrice()	Setter stores the new value	Returns 15.0f	PASSED
TC-ODT-10	toString()	Checks the text returned by toString()	Returns the formatted text	PASSED

3.14. ReviewTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-REV-01	postReview()	Posts a valid review into an empty list	List size is 1	PASSED
TC-REV-02	postReviewSetsDate()	Posts a review created without a date	Date is not null	PASSED

Test ID	Method Tested	Description	Expected Result	Status
TC-REV-03	postReviewInvalidScore()	Posts a review with score 6	Throws IllegalArgumentException and list stays empty	PASSED
TC-REV-04	postReviewEmptyComment()	Invalid input is rejected	Throws IllegalArgumentException	PASSED
TC-REV-05	postReviewWithoutBook()	Invalid input is rejected	Throws IllegalArgumentException	PASSED
TC-REV-06	postReviewWithoutCustomer()	Invalid input is rejected	Throws IllegalArgumentException	PASSED
TC-REV-07	getComment()	Getter returns the initialized value	Returns “Excelente libro”	PASSED
TC-REV-08	setScore()	Setter stores the new value	Returns 3	PASSED
TC-REV-09	setComment()	Setter stores the new value	Returns “Muy bueno”	PASSED
TC-REV-10	getBook()	Getter returns the initialized value	Returns the same object	PASSED
TC-REV-11	setScoreOutOfRangeIsRejected()	Sets a score of 9 (valid range 1-5)	Throws IllegalArgumentException	FAILED (DEF-REV-01)
TC-REV-12	postSameReviewTwiceDoesNotDuplicate()	Posts the same review twice into the list	List size is 1	FAILED (DEF-REV-02)

3.15. ShoppingCartTest.java

Test ID	Method Tested	Description	Expected Result	Status
TC-CRT-01	addBook()	Adds one book (quantity 2) to an empty cart	Items list size is 1	PASSED
TC-CRT-02	addBookSameBookTwice()	Adds the same book twice (2 and 3 units)	One item with quantity 5	PASSED
TC-CRT-03	calculateSubtotal()	Verifies: calculate subtotal	Returns 40.0f	PASSED
TC-CRT-04	removeBook()	Removes a book that is in the cart	Returns true and cart is empty	PASSED
TC-CRT-05	removeBookNotInCart()	Verifies: remove book not in cart	Returns false	PASSED
TC-CRT-06	clearShoppingCart()	Clears a cart that has one item	Items list is empty and subtotal is 0.0f	PASSED
TC-CRT-07	updateStockNotEnough()	Invalid input is rejected	Throws IllegalArgumentException	PASSED
TC-CRT-08	addBookSubtotal()	Verifies: add book subtotal	Returns 20.0f	PASSED
TC-CRT-09	getContentEmpty()	Getter returns the initialized value	Returns “The shopping cart is empty”	PASSED
TC-CRT-10	updateStock()	Verifies: update stock	Returns 7	PASSED
TC-CRT-11	removeNullBookReturnsFalse()	Removes null from a cart that has one item	Returns false	ERROR (DEF-CRT-01)